Kalman Filtering for Target Tracking and Indoor Positioning

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Code: `https://github.com/pengyu-jing/EE5103-Kalman-Filter-Project`

***Abstract*— This report designs, simulates and analyses Kalman filters for the three problems of the EE5103 project. For the scalar random-walk target of Part 1 the steady-state Riccati equation is solved in closed form ($K_f = 0.0951$, $P(k|k) = 0.0951$); a $10\,000$-sample simulation gives a bias of 0.0015 and a variance of 0.0945, i.e. a 10.4-fold reduction of the measurement-noise variance. For the constant-velocity target of Part 2 the filter is shown to converge to an α–β tracker with the exact gains $\alpha = 0.36$ and $\beta = 0.08$; the simulated position and velocity error variances (0.3586 and 0.0397) match the analytic covariance, and the larger full-range velocity variance is traced to the initial condition. For the indoor-positioning problem of Part 3 the centre and angular rate of the circular motion are identified from the measurements, and a system of two decoupled Kalman filters based on a harmonic-oscillator (constant-turn) model is proposed. A frequency-response analysis explains why a constant-velocity model enlarges the estimated circle by $\approx 20\%$, whereas the proposed filter has unit gain and zero phase at the motion frequency (internal-model principle); it passes the innovation consistency test and reduces the position RMSE by 48% in Monte Carlo tests.**

I. INTRODUCTION

The Kalman filter [1] is the minimum-variance state estimator for linear systems driven by Gaussian noise. It is the standard tool of target tracking [5] and of indoor positioning [10], where position fixes obtained from received signal strength are heavily corrupted by scattering, multipath and human movement. This project applies the filter derived in the course notes [2] to three problems of increasing difficulty: a target whose position is a random walk (Part 1), a target whose velocity is a random walk (Part 2), and a target moving on a circle that is observed through 25 noisy two-dimensional position fixes (Part 3).

Besides the requested graphs and statistics, the report explains the results analytically:

- closed-form steady-state solutions of the Riccati equation for Parts 1 and 2, which predict the simulated gains, covariances, variances and convergence rates;
- an explanation of the measured bias and variance, including the effect of the diffuse initialisation $P(0|{-}1) = 10^5 I$;
- for Part 3, a data-driven circular-motion model, a pair of decoupled oscillator-model Kalman filters, and a frequency-domain analysis that predicts the radius distortion, phase lag and RMSE of every design, validated by an innovation consistency test and Monte Carlo simulation.

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All simulations implement the filter equations of [2] directly (Python, NumPy/SciPy). Every number and table in this report is written by the simulation scripts (Appendix B).

II. KALMAN FILTER AND ANALYSIS TOOLS

A. Filter Equations

The process and the model used by the filter are

$$x(k+1) = Ax(k) + Bu(k) + w(k), \tag{1}$$

$$y(k) = Cx(k) + v(k), \tag{2}$$

where $w(k)$ and $v(k)$ are zero-mean, independent Gaussian noises with covariances R_1 and R_2. With the innovation $e(k) = y(k) - C\hat{x}(k|k-1)$ and its covariance $S(k) = CP(k|k-1)C^{\mathsf{T}} + R_2$, the Kalman filter is [2]

$$K_f(k) = P(k|k-1)C^{\mathsf{T}}S(k)^{-1}, \tag{3}$$

$$K(k) = AP(k|k-1)C^{\mathsf{T}}S(k)^{-1} = AK_f(k), \tag{4}$$

$$\hat{x}(k|k) = \hat{x}(k|k-1) + K_f(k)\,e(k), \tag{5}$$

$$\hat{x}(k+1|k) = A\hat{x}(k|k-1) + Bu(k) + K(k)\,e(k), \tag{6}$$

$$P(k|k) = P(k|k-1) - P(k|k-1)C^{\mathsf{T}}S(k)^{-1}CP(k|k-1), \tag{7}$$

$$P(k+1|k) = AP(k|k-1)A^{\mathsf{T}} - K(k)S(k)K^{\mathsf{T}}(k) + R_1. \tag{8}$$

In all three parts the filter is initialised as in the notes with $\hat{x}(0|-1) = 0$ and $P(0|-1) = 10^5 I$, i.e. with practically no prior information about $x(0)$.

B. Properties Used in the Analysis

P1 (error covariance). If $\hat{x}(0|-1)$ and $P(0|-1)$ are the mean and covariance of $x(0)$, the error $x(k) - \hat{x}(k|k)$ has zero mean and covariance $P(k|k)$, and $x(k+1) - \hat{x}(k+1|k)$ has covariance $P(k+1|k)$ [2, Example 4]. Bias and variance of the estimate are therefore predicted by 0 and $\operatorname{diag} P(k|k)$.

P2 (data independence). (3), (4), (7) and (8) do not contain $y(k)$: gains and covariances can be computed off-line, and filters that share $(A, C, R_1, R_2, P(0|-1))$ have identical gains and covariances.

P3 (convergence). If (A, C) is detectable and $(A, R_1^{1/2})$ is stabilisable, $P(k+1|k)$ converges, for any $P(0|-1) \geq 0$, to the unique stabilising solution $\bar{P}$ of the algebraic Riccati equation [4]

$$\bar{P} = A\bar{P}A^{\mathsf{T}} - A\bar{P}C^{\mathsf{T}}\big(C\bar{P}C^{\mathsf{T}} + R_2\big)^{-1}C\bar{P}A^{\mathsf{T}} + R_1, \tag{9}$$

and $A - \bar{K}C$ is stable. All models of this report satisfy these conditions.

P4 (steady-state filter). With constant gains $\bar{K}_f, \bar{K}$ the filter is linear time-invariant. From (5)–(6), the transfer function from y to $\hat{x}(k|k)$ is

$$H(z) = \big(I - \bar{K}_f C\big)\big(zI - A + \bar{K}C\big)^{-1}\bar{K} + \bar{K}_f. \quad (10)$$

A sinusoid $\cos(\omega k)$ at the input produces $|H(e^{j\omega})|\cos(\omega k + \angle H(e^{j\omega}))$ at the output, and white noise of variance R_2 produces an output variance $R_2\|H\|_2^2$, where $\|H\|_2^2 = \sum_{i\geq 0} h(i)^2$ is the squared H_2 norm of the impulse response $h(i)$.

C. Performance Statistics

For a run of $N+1$ samples the project defines the bias and the variance of the estimation error as

$$b = \frac{1}{N+1}\sum_{k=0}^{N}\big(x(k) - \hat{x}(k|k)\big),$$
$$V = \frac{1}{N+1}\sum_{k=0}^{N}\big(x(k) - \hat{x}(k|k)\big)^2. \quad (11)$$

V is the mean squared error ($V = b^2 +$ sample variance); for an unbiased filter in steady state it estimates $\bar{P}(k|k)$.

III. PART 1: RANDOM-WALK TARGET

A. Model and Filter

The target obeys $x(k+1) = x(k) + w(k)$, $y(k) = x(k) + v(k)$ with $\sigma_w = 0.1$, $\sigma_v = 1$ and $x(0) = 5$. The filter uses the same model: $A = C = 1$, $B = 0$, $R_1 = \sigma_w^2 = 0.01$, $R_2 = \sigma_v^2 = 1$. One run of $k = 0, \ldots, N$ with $N = 10\,000$ is simulated.

B. Analytic Steady State

With $A = C = 1$, (8) reduces to the scalar recursion

$$P(k+1|k) = \frac{R_2\,P(k|k-1)}{P(k|k-1) + R_2} + R_1. \quad (12)$$

At steady state $P(k+1|k) = P(k|k-1) = \bar{P}$, which gives $\bar{P}^2 - R_1\bar{P} - R_1R_2 = 0$ and

$$\bar{P} = \tfrac{1}{2}\Big(R_1 + \sqrt{R_1^2 + 4R_1R_2}\Big) = 0.1051, \quad (13)$$

$$\bar{K}_f = \frac{\bar{P}}{\bar{P} + R_2} = 0.0951, \quad \bar{P}(k|k) = R_2\bar{K}_f = 0.0951, \quad (14)$$

and $\bar{K} = A\bar{K}_f = 0.0951$. Dividing by R_2 shows that the gain depends only on the ratio $q = R_1/R_2$: $\bar{K}_f = \frac{1}{2}(-q + \sqrt{q^2 + 4q})$. Moreover, (3) and (7) give the exact identity $P(k|k) = R_2K_f(k)/C$ for *every* k [2]. With $R_2 = C = 1$, Graph 2 and Graph 3 are therefore the same curve, and $K(k) = AK_f(k) = K_f(k)$.

C. Results

Graph 1 is shown in Fig. 1, Graphs 2 and 3 in Fig. 2, the bias and variance in Table I, and a Monte Carlo check of $P(k|k)$ in Fig. 3. The estimate follows the slowly drifting random walk closely while rejecting most of the measurement noise (Fig. 1b). $P(k|k)$ and $K_f(k)$ converge to the analytic values (14); at $k = N$ the simulation gives $P(N|N) = K_f(N) = 0.0951$.

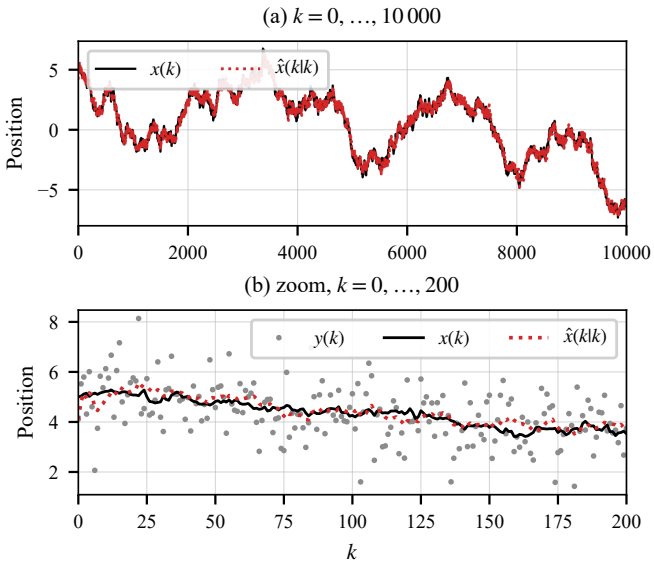


Fig. 1. Part 1, Graph 1: true state $x(k)$ (solid) and estimate $\hat{x}(k|k)$ (dotted) for (a) the whole run and (b) the first 200 samples, with the measurements $y(k)$.

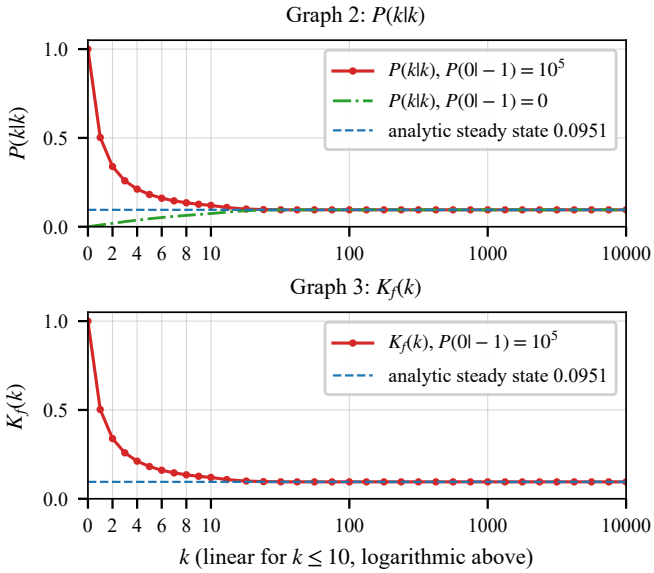


Fig. 2. Part 1, Graph 2: $P(k|k)$ and Graph 3: $K_f(k)$. The dash-dot curve uses the known initial state ($\hat{x}(0|-1) = 5$, $P(0|-1) = 0$). Dashed: analytic steady state (14).

D. Discussion

1) Transient: from averaging to forgetting. With the diffuse prior, $K_f(0) = 0.99999$ and $P(0|0) = 0.99999 \approx R_2$: the first estimate is simply the first measurement. While $P(k|k-1) \gg R_1$ the random walk is negligible and the filter computes the recursive sample mean, $K_f(k) \approx 1/(k+1)$; indeed $K_f(1) = 0.5025$ and $K_f(2) = 0.3388$ (compare Example 3 of [2], where $R_1 = 0$ gives the plain average). Once the averaging window becomes comparable with the time over which x drifts, the process noise stops the decrease, and $K_f(k)$ is within 1% of $\bar{K}_f$ after $k = 26$ samples. Starting instead from the known initial state ($P(0|-1) = 0$), $P(k|k)$

TABLE I

BIAS AND VARIANCE (11) OF THE ESTIMATION ERROR (PARTS 1 AND 2)

	single run $k = 0..N$	$k \geq 100$	100 runs, $k = 0..N$ mean $\pm$ std	theory
Part 1				
bias	0.0015	0.0020	-0.0028 ± 0.0131	0 ± 0.0138
variance	0.0945	0.0948	0.0952 ± 0.0047	0.0951
Part 2				
bias x_1	0.0093	0.0083	0.0008 ± 0.0133	0
bias x_2	0.0083	0.0051	0.0035 ± 0.0045	0
variance x_1	0.3578	0.3586	0.3612 ± 0.0090	0.3600
variance x_2	0.1297	0.0397	0.1303 ± 0.0010	0.0400 (0.13†)

†Including the $k = 0$ term $30^2/(N+1) = 0.0900$, see Section IV-D.

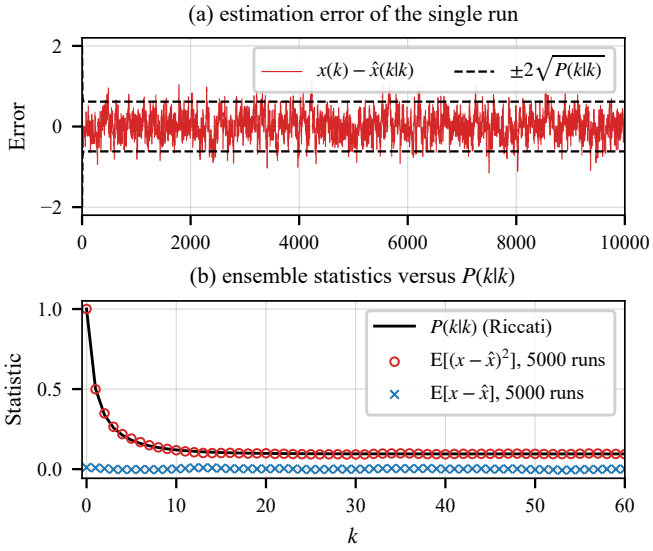


Fig. 3. Part 1: (a) estimation error of the single run with the $\pm 2\sqrt{P(k|k)}$ band; (b) mean and mean-square error over 5000 independent runs compared with $P(k|k)$.

grows from zero and reaches the *same* limit after 27 samples (Fig. 2), as predicted by P3.

2) Steady-state filter. With constant gain, (5) becomes the exponential smoother

$$\hat{x}(k|k) = (1 - \bar{K}_f)\,\hat{x}(k-1|k-1) + \bar{K}_f\, y(k), \qquad (15)$$

a first-order low-pass filter with pole $1 - \bar{K}_f = 0.9049$, time constant $-1/\ln(1 - \bar{K}_f) = 10.0$ samples and an equivalent averaging window of $(2 - \bar{K}_f)/\bar{K}_f = 20$ samples.

3) Optimal compromise between lag and noise. With $a = 1 - \bar{K}_f$ the steady-state error $\tilde{x}(k) = x(k) - \hat{x}(k|k)$ obeys $\tilde{x}(k) = a[\tilde{x}(k-1) + w(k-1)] - \bar{K}_f v(k)$, hence

$$\mathrm{E}[\tilde{x}^2] = \underbrace{\frac{a^2 R_1}{1 - a^2}}_{\text{lag: } 0.0452} + \underbrace{\frac{\bar{K}_f^2 R_2}{1 - a^2}}_{\text{noise: } 0.0499} = 0.0951. \qquad (16)$$

A larger gain shortens the lag behind the drifting target but passes more measurement noise; a smaller gain does the opposite. Minimising (16) over the gain returns exactly $\bar{K}_f = 0.0951$: the Kalman gain is the optimal compromise, and the two error sources are almost equal at the optimum.

4) Bias and variance. The bias $b = 0.0015$ is statistically zero. Because $\tilde{x}(k)$ is a first-order autoregressive sequence

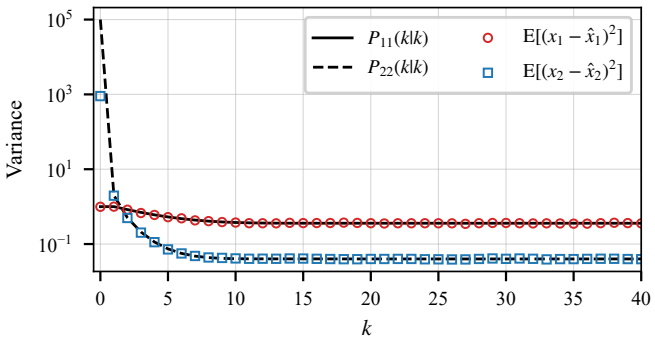


Fig. 4. Part 2: diagonal of $P(k|k)$ versus the mean-square errors over 5000 runs.

with coefficient a, the time average of $N+1$ samples has a standard deviation of $\sqrt{P(k|k)(1+a)/((1-a)(N+1))} = 0.0138$ (the correlation inflates the variance 20.0 times compared with independent samples), which agrees with the spread 0.0131 observed over 100 independent runs. The variance $V = 0.0945$ agrees with $\bar{P}(k|k) = 0.0951$ within the run-to-run spread (± 0.0047). Compared with the raw measurement error variance (0.9796) the filter reduces the error variance 10.4-fold (10.2 dB), and 95.2% of the errors lie inside the $\pm 2\sqrt{P(k|k)}$ band (Gaussian theory: 95.4%, Fig. 3a).

5) Ensemble check. Over 5000 runs the mean error is zero and the mean-square error equals $P(k|k)$ at every k, including the transient (Fig. 3b; largest relative deviation 6.5% for $k \geq 1$, consistent with the $\approx 2\%$ sampling error of a 5000-run average). This confirms P1: $P(k|k)$ is the error variance, not just an internal filter variable.

IV. PART 2: CONSTANT-VELOCITY TARGET

A. Model and Filter

The target obeys (5.3)–(5.4) of the project with $T = 1$, $\sigma_w = 0.1$, $\sigma_v = 1$ and $x(0) = [0 \;\; 30]^{\mathsf{T}}$:

$$A = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} \frac{1}{2}T^2 \\ T \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad (17)$$

with $x(k+1) = Ax(k) + Gw(k)$. The process-noise covariance used by the filter is

$$R_1 = \mathrm{E}\big[Gw(Gw)^{\mathsf{T}}\big] = \sigma_w^2 \begin{bmatrix} \frac{T^4}{4} & \frac{T^3}{2} \\ \frac{T^3}{2} & T^2 \end{bmatrix} = \begin{bmatrix} 0.0025 & 0.005 \\ 0.005 & 0.01 \end{bmatrix}, \qquad (18)$$

and $R_2 = \sigma_v^2 = 1$.

B. Analytic Steady State

Let $\bar{P} = [m_{ij}]$ be the steady value of $P(k+1|k)$, $S = m_{11} + R_2$, and write the steady gain as $\bar{K}_f = [\alpha \;\; \beta/T]^{\mathsf{T}}$, i.e. $\alpha = m_{11}/S$ and $\beta = T m_{12}/S$. Then $R_2 = S - m_{11} = S(1-\alpha)$ and (7) gives

$$\bar{P}(k|k) = \begin{bmatrix} \alpha\sigma_v^2 & \beta\sigma_v^2/T \\ \beta\sigma_v^2/T & m_{22} - m_{12}^2/S \end{bmatrix}. \qquad (19)$$

Substituting (19) into $\bar{P} = A\bar{P}(k|k)A^{\mathsf{T}} + R_1$, the $(2,2)$ entry gives $m_{12}^2/S = \sigma_w^2 T^2$, i.e.

$$\frac{\beta^2}{1-\alpha} = \lambda^2, \qquad \lambda = \frac{\sigma_w T^2}{\sigma_v}, \tag{20}$$

where λ is the tracking index [6]. Eliminating m_{22} between the $(1,1)$ and $(1,2)$ entries gives $(\alpha - \beta/2)^2 = 2\beta(1-\alpha)$. With $u = \sqrt{1-\alpha}$ this is $\beta = 2(1-u)^2$, and (20) becomes $2u^2 - (4+\lambda)u + 2 = 0$, whose root in $(0,1)$ is

$$u = \tfrac{1}{4}\Big(4 + \lambda - \sqrt{\lambda^2 + 8\lambda}\Big), \quad \alpha = 1 - u^2, \quad \beta = \lambda u. \tag{21}$$

Here $\lambda = 0.10$, $\sqrt{\lambda^2 + 8\lambda} = 0.9$, so $u = 0.80$, $\alpha = 0.36$ and $\beta = 0.08$ *exactly*, and

$$\begin{aligned} \bar{P}(k|k) &= \begin{bmatrix} 0.3600 & 0.0800 \\ 0.0800 & 0.0400 \end{bmatrix}, \quad \bar{K}_f = \begin{bmatrix} 0.36 \\ 0.08 \end{bmatrix}, \\ \bar{P}(k+1|k) &= \begin{bmatrix} 0.5625 & 0.1250 \\ 0.1250 & 0.0500 \end{bmatrix}, \quad \bar{K} = \begin{bmatrix} 0.44 \\ 0.08 \end{bmatrix}, \end{aligned} \tag{22}$$

with $\bar{P}_{22}(k|k) = \sigma_v^2 \beta(\alpha - \beta/2)/((1-\alpha)T^2)$. These values coincide with the numerical solution of (9) to machine precision. The steady-state Kalman filter is the classical α–β tracker [5].

C. Results

Graphs 4–5 are shown in Fig. 5, Graphs 6–9 in Fig. 6, the statistics in Table I and a Monte Carlo check in Fig. 4. The position grows to $x_1(N) = 319272$, so at the scale of Graph 4 true and estimated positions are indistinguishable; the error itself (Fig. 5b) stays inside $\pm 2\sqrt{P_{11}(k|k)}$. The velocity wanders between 18 and 47 and is tracked closely after the first two samples (Fig. 5c,d). All covariances and gains converge to (22); at $k = N$ the simulation gives $P_{11} = 0.3600$, $P_{12} = 0.0800$, $P_{22} = 0.0400$, $K_{f1} = 0.3600$ and $K_{f2} = 0.0800$.

D. Discussion

1) Start-up. At $k = 0$, $K_f(0) = [1.0000 \quad 0.0000]^{\mathsf{T}}$: the first measurement fixes the position ($P_{11}(0|0) = 1$) but contains no information about the velocity ($P_{22}(0|0) \approx 10^5$, $\hat{x}_2(0|0) = 0$). At $k = 1$, $K_f(1) = [1.0000 \quad 1.0000]^{\mathsf{T}}$, so $\hat{x}_1(1|1) = y(1)$ and $\hat{x}_2(1|1) = y(1) - y(0)$: the filter starts with the two-point difference, whose error $\frac{1}{2}w(0) - v(1) + v(0)$ has variance $2\sigma_v^2 + \sigma_w^2/4 = 2.0025 = P_{22}(1|1)$. Two samples are needed because (A, C) has observability index 2. Afterwards the filter averages over more and more samples, and all gains are within 1% of (22) after $k = 13$.

2) Dynamics of the steady-state filter. The eigenvalues of $A - \bar{K}C$ are $0.7800 \pm j0.1778$, with modulus $0.8000 = u = \sqrt{1-\alpha}$ and time constant 4.5 samples. The filter is about twice as fast as in Part 1 (pole 0.9049).

3) Interpretation of $\bar{P}(k|k)$. Although σ_w and σ_v are the same as in Part 1, the position error variance is larger (0.3600 against 0.0951). In Part 2 the noise drives the velocity, so the position uncertainty of a coasting target grows like $k^{3/2}$ instead of $k^{1/2}$. The filter must therefore forget old data faster ($\alpha = 0.36$ against $\bar{K}_f = 0.0951$), which lets more measurement noise through. The position error is still 2.8 times smaller than the raw measurement noise. The velocity, which is never measured, is estimated with variance 0.0400 (standard deviation 0.2); naive differencing $y(k) - y(k-1)$ gives 1.959 in the same run, which is 49 times worse. The positive covariance $\bar{P}_{12} = 0.0800$ means that a too-large velocity estimate tends to come with a too-large position estimate, as the model propagates one into the other.

4) Bias and variance. Both biases are statistically zero (Table I), and the position variance 0.3578 matches $\bar{P}_{11} = 0.3600$. The velocity variance over $k = 0, \ldots, N$ is 0.1297, more than three times $\bar{P}_{22}$. The excess comes entirely from the single term $k = 0$: $\hat{x}_2(0|0) = 0$ while $x_2(0) = 30$, so this term contributes $30^2/(N+1) = 0.0900$ to V (and $30/(N+1) = 0.003$ to b). Without it, $V = 0.0397 \approx \bar{P}_{22}$. This is not a filter error: $P_{22}(0|0) \approx 10^5$ correctly reports that the velocity is unknown at $k = 0$. Because (11) averages over the whole run, the statistic depends on the initialisation. Excluding the transient ($k \geq 100$) gives 0.3586 and 0.0397, in agreement with (22).

5) Ensemble check. The mean-square errors over 5000 runs coincide with $P_{11}(k|k)$ and $P_{22}(k|k)$ for all $k \geq 1$ (largest deviation 3.8%, Fig. 4). At $k = 0$ the ensemble velocity error is $900 = 30^2$, far below $P_{22}(0|0) \approx 10^5$. P1 holds only if the prior describes $x(0)$ correctly. $P(0|-1) = 10^5 I$ is a deliberately conservative prior, and after two samples it has been forgotten.

6) Dependence on the noise ratio. The closed forms (14) and (21) show that the steady-state filters of Parts 1 and 2 depend on the noise levels only through $q = R_1/R_2$ and the tracking index λ (Fig. 7). For $q \ll 1$, $\bar{K}_f \approx \sqrt{q}$ (0.1000 against the exact 0.0951), so $\bar{P}(k|k) = R_2 \bar{K}_f \approx \sigma_w \sigma_v$. The accuracy is therefore set by the geometric mean of the two noise levels, and the bandwidth of the filter, $1 - (1 - \bar{K}_f) \approx \sqrt{q}$, grows only with the square root of the noise ratio. For $\lambda \ll 1$, $\alpha \approx \sqrt{2\lambda}$ and $\beta \approx \lambda$; at $\lambda = 0.10$ these approximations (0.447 and 0.10) are only indicative, which is why the exact solution (21) was used. When the process noise dominates ($q \gg 1$, $\lambda \gg 1$), $\bar{K}_f \to 1$ and $\alpha \to 1$: the filter no longer smooths and simply returns the latest position measurement, as in Example 3 of [2] with $R_2 = 0$.

V. PART 3: INDOOR POSITIONING OF A TARGET IN CIRCULAR MOTION

A. Identification of the Circular Motion

A target in uniform circular motion satisfies

$$y(t) = c_y + r\cos(\omega t + \theta_0), \qquad z(t) = c_z + r\sin(\omega t + \theta_0). \tag{23}$$

The parameters are identified from the 25 measurements by least squares. (i) Circle fit [8]: minimising $\sum_k \left(y^2 + z^2 + Dy + Ez + F\right)^2$, which is linear in (D, E, F), gives $\hat{c} = (-D/2, -E/2) = (0.33, 0.28)$ and $\hat{r} = \sqrt{\hat{c}_y^2 + \hat{c}_z^2 - F} = 10.24$. (ii) Angular rate: fitting the unwrapped bearing $\theta(k) = \operatorname{atan2}(z - \hat{c}_z, y - \hat{c}_y) \approx \theta_0 + \omega k$ gives $\hat{\omega} = 0.2616 \pm 0.0032$ rad/sample (14.99°/sample, period 24.02 samples),

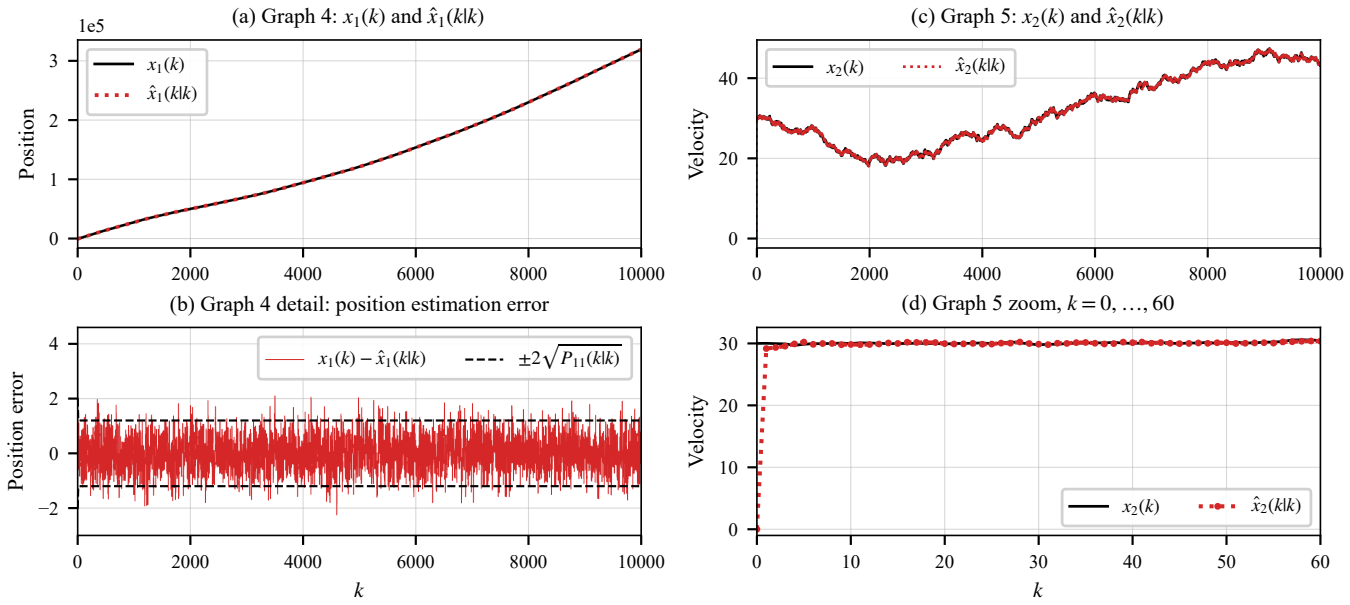


Fig. 5. Part 2, Graph 4: (a) position $x_1(k)$ (solid) and $\hat{x}_1(k|k)$ (dotted); (b) position error with the $\pm 2\sqrt{P_{11}(k|k)}$ band (the error is invisible at the scale of (a)). Graph 5: (c) velocity $x_2(k)$ (solid) and $\hat{x}_2(k|k)$ (dotted); (d) the first 60 samples.

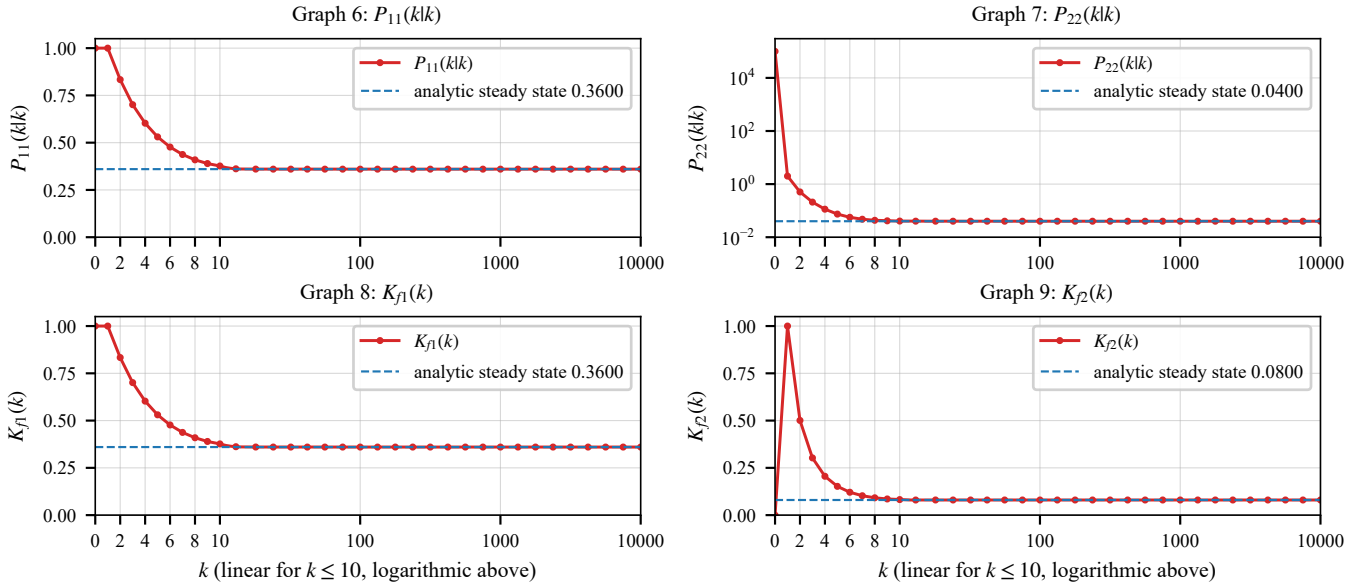


Fig. 6. Part 2, Graphs 6–9: $P_{11}(k|k)$, $P_{22}(k|k)$ (logarithmic scale), $K_{f1}(k)$ and $K_{f2}(k)$. Dashed: analytic steady state (22).

i.e. one revolution over the 24 sampling intervals, starting at $\hat{\theta}_0 = 0.08°$. The centripetal acceleration is $\hat{r}\hat{\omega}^2 = 0.70$ per sample^2. The standard errors of $\hat{c}_y$ and $\hat{c}_z$ (from the Monte Carlo study below) are 0.28 and 0.29, so the centre is not significantly different from the origin. The identified offset is kept anyway, which keeps the design general.

B. The System of Kalman Filters

The system (Fig. 8) consists of two Kalman filters, one per coordinate, whose inputs are $y(k)$ and $z(k)$ and whose outputs are $\hat{x}_y(k|k)$ and $\hat{x}_z(k|k)$. The filters can be decoupled because the two measurement noises are independent and, by (23), each coordinate obeys the same second-order equation $\ddot{p} = -\omega^2(p - c)$ on its own; the two axes are linked only through their initial conditions (a 90° phase shift). Both filters use the same A, C, R_1, $R_2 = 1$ and $P(0|-1)$, so by P2 their gains and covariances are identical and are reported once.

Two models were designed.

Model M1 (baseline, constant velocity). Each axis uses the model of Part 2, (17)–(18), with state $[p \;\; \dot{p}]^\mathsf{T}$. On a circle the acceleration has magnitude $\hat{r}\hat{\omega}^2 = 0.70$, and the constant-velocity model can only treat it as process noise. Following the rule $0.5a_M \leq \sigma_w \leq a_M$ for the maximum acceleration a_M [5], $\sigma_w = 0.7 \approx a_M$ is used. The Part 2 value $\sigma_w = 0.1$

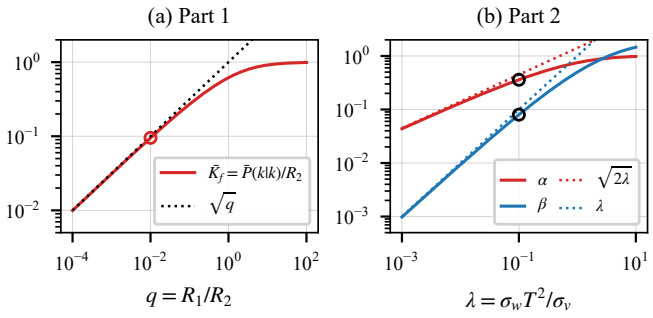


Fig. 7. Steady-state gains versus the noise ratio: (a) Part 1, $\bar{K}_f(q)$ from (14); (b) Part 2, $\alpha(\lambda)$ and $\beta(\lambda)$ from (21). Dotted: asymptotes for small ratios; circles: the values of this project.

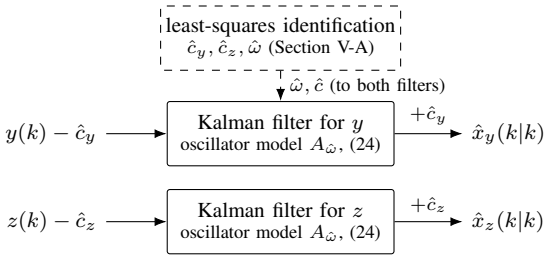


Fig. 8. The system of Kalman filters of Part 3. The parameters identified in Section V-A set $A_{\hat{\omega}}$ and the centre offsets; the y- and z-filters are identical.

is included for comparison.

Model M2 (proposed, harmonic oscillator / constant turn). Each axis uses the state $x = [p - \hat{c} \;\; \dot{p}]^\mathsf{T}$ with $\dot{x} = \left[\begin{smallmatrix} 0 & 1 \\ -\omega^2 & 0 \end{smallmatrix}\right] x$, whose exact (zero-order-hold) discretisation [3] is

$$A_{\hat{\omega}} = \begin{bmatrix} \cos\hat{\omega}T & \frac{\sin\hat{\omega}T}{\hat{\omega}} \\ -\hat{\omega}\sin\hat{\omega}T & \cos\hat{\omega}T \end{bmatrix} = \begin{bmatrix} 0.9660 & 0.9886 \\ -0.0676 & 0.9660 \end{bmatrix}, \tag{24}$$

with G and C as in (17), $R_1 = \sigma_w^2 GG^\mathsf{T}$, $\sigma_w = 0.1$ and $R_2 = 1$. The eigenvalues of $A_{\hat{\omega}}$ are $e^{\pm j\hat{\omega}T}$, so the noise-free model generates exactly the sinusoids of (23). The small process noise absorbs identification errors and deviations from a perfect circle. The outputs are $\hat{x}_y(k|k) = \hat{c}_y + \hat{x}_1(k|k)$ and $\hat{x}_z(k|k) = \hat{c}_z + \hat{x}_1(k|k)$. In the original coordinates the offset is an input term $Bu = (I - A_{\hat{\omega}})[\hat{c} \;\; 0]^\mathsf{T}$ in (1) and (6). Both models are initialised with $\hat{x}(0|-1) = 0$ and $P(0|-1) = 10^5 I$.

C. Results

Fig. 9 is Figure 5.1 of the project, redrawn from the tabulated measurements, with the estimates of the final design (M2) superimposed. The requested $K_f(k)$, $K(k)$, $P(k|k)$ and $P(k+1|k)$ of the final design are listed in Table II. The same quantities for the baseline M1 and the numerical estimates are given in Appendix A. Fig. 10 compares the three designs, Table III summarises their performance, Fig. 11 shows a Monte Carlo evaluation and Fig. 12 the influence of σ_w.

D. Discussion

1) Gains and covariances. At $k = 0$, $K_f(0) = [1 \;\; 0]^\mathsf{T}$ and the estimate equals the first measurement, while $K(0) = A_{\hat{\omega}} K_f(0) = [\cos\hat{\omega} \;\; -\hat{\omega}\sin\hat{\omega}]^\mathsf{T} = [0.9660 \;\; -0.0676]^\mathsf{T}$ rotates this fix along the circle for the prediction $\hat{x}(1|0)$. At $k = 1$ the velocity is obtained from the two fixes ($K_{f2}(1) \approx 1$). Afterwards $K_{f1}(k)$ decreases to 0.2785 and $P_{11}(k|k)$ to 0.278 ($P_{11}(24|24) = 0.279$, Table II). The baseline M1 settles after only about ten samples at the much larger gain $K_{f1} = 0.6886$ (Table IV): because its model cannot predict the turning motion, it has to rely on the measurements.

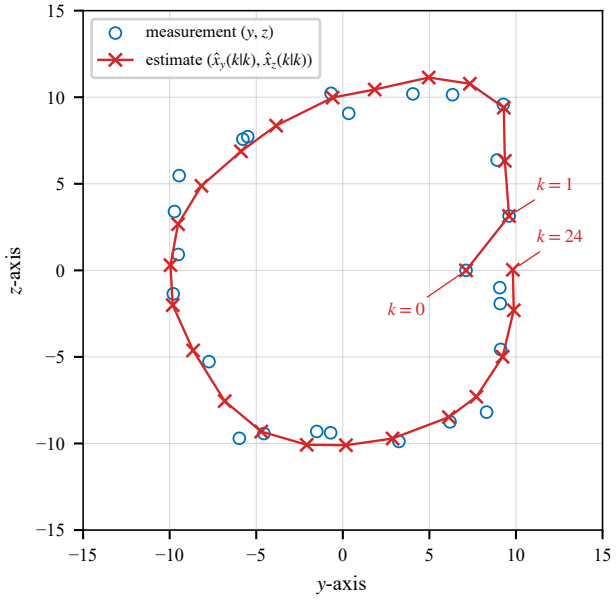


Fig. 9. Figure 5.1 of the project (measurements ○) with the estimates $(\hat{x}_y(k|k), \hat{x}_z(k|k))$ of the final design (model M2) superimposed as × and joined by a solid line.

2) Frequency-domain interpretation. In steady state both coordinates pass through the same $H(z)$ (10), and the noise-free inputs are $\hat{r}\cos(\hat{\omega}k + \theta_0)$ and $\hat{r}\sin(\hat{\omega}k + \theta_0)$ plus a constant (which passes with unit gain, $H(1) = 1$). The estimated trajectory is therefore again a circle, of radius $|H(e^{j\hat{\omega}})|\hat{r}$, lagging by $-\angle H(e^{j\hat{\omega}})$. For M1 with the Part 2 tuning ($\sigma_w = 0.1$), $|H| = 1.197$ and $\angle H = -21.5°$. The predicted radius 12.26 matches the mean radius 12.11 of the estimates in Fig. 10a, and the lag is 1.4 samples. The cause is resonance: the steady-state α–β filter (identical to that of Part 2) is a lightly damped second-order system whose poles ($0.7800 \pm j0.1778$) have an angle of 12.8°/sample, close to the target's 15°/sample. With $\sigma_w = 0.7$ the bandwidth is larger ($|H| = 1.049$, $\angle H = -1.4°$), but the noise gain $R_2\|H\|_2^2$ grows from 0.289 to 0.600. For the oscillator design, $H(e^{j\hat{\omega}}) = 1$ exactly. This is the internal-model principle [9]: if the measurements follow the model without noise, the prediction error obeys $\varepsilon(k+1) = (A_{\hat{\omega}} - \bar{K}C)\varepsilon(k) \to 0$, so the circular motion is reproduced with unit gain and zero phase, while the noise gain is only 0.187. For a circle in white noise the asymptotic RMSE of the two-dimensional position follows from P4:

$$\mathrm{RMSE}_\infty^2 = 2R_2\|H\|_2^2 + \hat{r}^2\left|H(e^{j\hat{\omega}}) - 1\right|^2, \tag{25}$$

the sum of the noise passed by both filters and the squared

TABLE II

PART 3, FINAL DESIGN (OSCILLATOR MODEL, $\sigma_w = 0.1$): $K_f(k)$, $K(k)$, $P(k|k)$ AND $P(k+1|k)$. THE y- AND z-FILTERS HAVE IDENTICAL VALUES. THE ROW ∞ IS THE STEADY-STATE (RICCATI) SOLUTION.

	$K_f(k)$		$K(k)$		$P(k\|k)$			$P(k+1\|k)$		
k	K_{f1}	K_{f2}	K_1	K_2	P_{11}	P_{12}	P_{22}	P_{11}	P_{12}	P_{22}
0	1.0000	0.0000	0.9660	−0.0676	1.0000	0.0000	1.000×10^5	9.774×10^4	9.550×10^4	9.331×10^4
1	1.0000	0.9771	1.9319	0.8762	1.0000	0.9771	1.9804	4.7375	2.6774	1.7349
2	0.8257	0.4666	1.2590	0.3949	0.8257	0.4666	0.4855	2.1388	0.8189	0.4058
3	0.6814	0.2609	0.9162	0.2059	0.6814	0.2609	0.1922	1.3245	0.3700	0.1583
4	0.5698	0.1592	0.7078	0.1152	0.5698	0.1592	0.0994	0.9354	0.2006	0.0846
5	0.4833	0.1037	0.5694	0.0674	0.4833	0.1037	0.0638	0.7139	0.1241	0.0582
6	0.4165	0.0724	0.4740	0.0418	0.4165	0.0724	0.0492	0.5776	0.0875	0.0484
7	0.3661	0.0555	0.4085	0.0288	0.3661	0.0555	0.0435	0.4926	0.0707	0.0450
8	0.3300	0.0474	0.3656	0.0234	0.3300	0.0474	0.0417	0.4417	0.0643	0.0442
9	0.3064	0.0446	0.3400	0.0223	0.3064	0.0446	0.0413	0.4139	0.0631	0.0442
10	0.2927	0.0446	0.3269	0.0233	0.2927	0.0446	0.0413	0.4013	0.0640	0.0441
11	0.2864	0.0457	0.3218	0.0247	0.2864	0.0457	0.0412	0.3972	0.0652	0.0438
12	0.2843	0.0466	0.3207	0.0258	0.2843	0.0466	0.0407	0.3966	0.0657	0.0432
13	0.2840	0.0470	0.3209	0.0262	0.2840	0.0470	0.0401	0.3966	0.0655	0.0426
14	0.2840	0.0469	0.3207	0.0261	0.2840	0.0469	0.0395	0.3957	0.0648	0.0420
15	0.2835	0.0464	0.3198	0.0257	0.2835	0.0464	0.0390	0.3939	0.0640	0.0416
16	0.2826	0.0459	0.3183	0.0252	0.2826	0.0459	0.0387	0.3917	0.0633	0.0414
17	0.2814	0.0455	0.3168	0.0249	0.2814	0.0455	0.0385	0.3896	0.0628	0.0413
18	0.2804	0.0452	0.3155	0.0247	0.2804	0.0452	0.0385	0.3880	0.0626	0.0413
19	0.2796	0.0451	0.3146	0.0246	0.2796	0.0451	0.0385	0.3871	0.0625	0.0413
20	0.2790	0.0451	0.3141	0.0247	0.2790	0.0451	0.0385	0.3866	0.0625	0.0413
21	0.2788	0.0451	0.3139	0.0247	0.2788	0.0451	0.0385	0.3864	0.0626	0.0413
22	0.2787	0.0451	0.3138	0.0248	0.2787	0.0451	0.0384	0.3863	0.0626	0.0412
23	0.2787	0.0452	0.3138	0.0248	0.2787	0.0452	0.0384	0.3863	0.0626	0.0412
24	0.2787	0.0452	0.3138	0.0248	0.2787	0.0452	0.0384	0.3863	0.0626	0.0412
∞	0.2785	0.0451	0.3136	0.0247	0.2785	0.0451	0.0384	0.3860	0.0625	0.0412

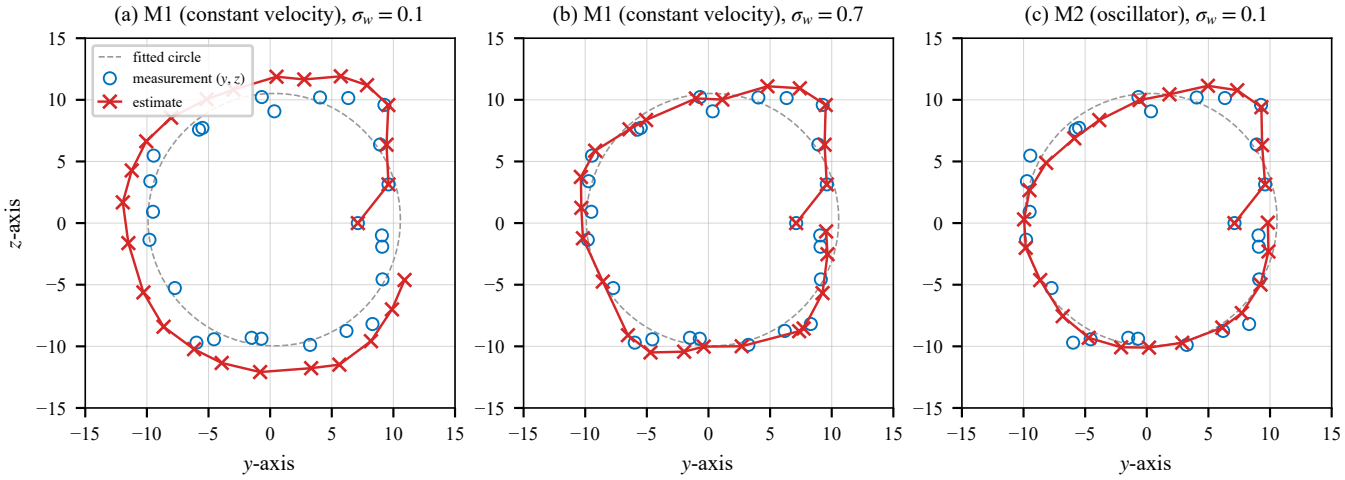


Fig. 10. Part 3: estimates of (a) M1 with $\sigma_w = 0.1$, (b) M1 with $\sigma_w = 0.7$ and (c) the proposed M2. Dashed: identified circle.

distance between the true and the distorted circle ($\hat{r}|H - 1| = 4.64$ and 0.56 for the two M1 designs). Equation (25) predicts 4.70, 1.23 and 0.61, against the Monte Carlo values 4.67, 1.23 and 0.73 at $k = 24$ (Table III). The remaining gap for M2 is caused by identifying $\hat{c}$ and $\hat{\omega}$ from only 25 noisy samples: with the true parameters the Monte Carlo RMSE is 0.60. Note that P_{11} of M1 (0.360 for $\sigma_w = 0.1$) is the error variance the filter *believes* in under its own model. The actual error is far larger, because the true acceleration is not white noise.

3) Consistency test on the measured data. If the model is correct, the normalised innovation squared $e(k)^2/S(k)$ (Section II) has mean one [5]. Averaged over $k = 2, \ldots, 24$ and both axes (46 terms), it must lie in $[0.63, 1.45]$ at the 95% level. M1 with $\sigma_w = 0.1$ gives 12.38: its innovations are about 3.5 times larger than the filter expects, so this design is clearly rejected. M1 with $\sigma_w = 0.7$ gives 1.40 (borderline), and M2 gives 1.23 (consistent). Unlike the RMSE, this test needs no knowledge of the true trajectory.

4) Monte Carlo evaluation. Synthetic circles with the identified geometry and unit-variance noise were filtered 5000 times, with the identification repeated on every run (Fig. 11). Over $k = 5, \ldots, 24$ the position RMSE is 1.41 for the raw measurements, 4.03 for M1 with $\sigma_w = 0.1$, 1.23 for

TABLE III

COMPARISON OF THE PART 3 DESIGNS

	M1 $\sigma_w = 0.1$	M1 $\sigma_w = 0.7$	M2 $\sigma_w = 0.1$	raw
P_{11} (Riccati)	0.360	0.689	0.278	1.000
$R_2 \|H\|_2^2$	0.289	0.600	0.187	1.000
$\|H(e^{j\hat{\omega}})\|$	1.197	1.049	1.000	1
$\angle H(e^{j\hat{\omega}})$	$-21.5°$	$-1.4°$	$0.0°$	$0°$
radius $\hat{r}\|H(e^{j\hat{\omega}})\|$	12.26	10.74	10.24	10.24
mean radius of estimates	12.11	10.65	10.21	10.14
RMS radial deviation	1.93	0.85	0.60	0.83
mean NIS	12.38	1.40	1.23	–
RMSE, $k = 5, \ldots, 24$	4.03	1.23	0.74	1.41
RMSE, $k = 24$	4.67	1.23	0.73	1.41
RMSE_∞, theory (25)	4.70	1.23	0.61	1.41

Rows 1–5: steady-state analysis at $\hat{\omega}$. Rows 6–8: measured data (radius for $k \geq 5$; NIS for $k \geq 2$, 95% region $[0.63, 1.45]$). Rows 9–11: position RMSE, 5000 Monte Carlo runs on a synthetic circle with the identified geometry.

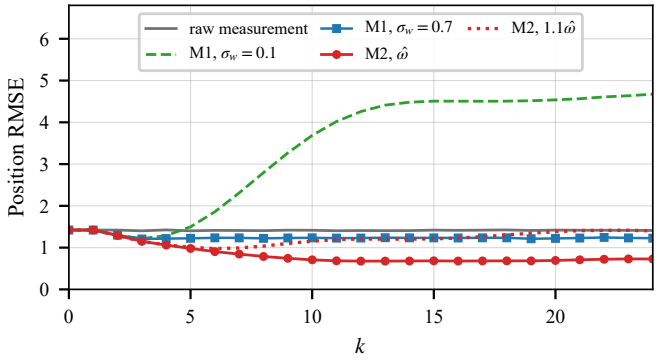


Fig. 11. Part 3: position RMSE over 5000 synthetic runs with the identified geometry. The oscillator design re-identifies $\hat{\omega}$ and $\hat{c}$ in every run; the dotted curve uses an angular rate that is 10% too large.

M1 with $\sigma_w = 0.7$ (13% better than raw) and 0.74 for M2 (48% better than raw). If the angular rate is overestimated by 10%, the RMSE of M2 rises to 1.23, so the accuracy of $\hat{\omega}$ (standard error 1.3% here) matters.

5) Choice of σ_w. Fig. 12 repeats the analysis for σ_w between 0.003 and 3. For M1 the asymptotic RMSE (25) has a minimum of 1.21 at $\sigma_w \approx 1.00$: a smaller σ_w distorts the circle, and a larger one passes more noise. The NIS test accepts M1 only for $\sigma_w \gtrsim 0.68$. The chosen $\sigma_w = 0.7$ is therefore consistent and within 2% of the best RMSE the constant-velocity model can reach, yet even this optimum improves on the raw measurements by only 15%. For M2 the NIS lies inside the 95% region for every σ_w between 0.003 and 1.26. Equation (25) decreases towards zero as $\sigma_w \to 0$ because it assumes the true ω and an infinitely long record. With 25 samples and an identified ω, the Monte Carlo RMSE levels off at 0.71 for $\sigma_w \leq 0.01$ and is 0.74 at $\sigma_w = 0.1$. This small gain is not worth the lower tolerance of a smaller σ_w to deviations from a perfect circle; indeed the measured data are explained slightly better by $\sigma_w = 0.1$ (NIS 1.23) than by $\sigma_w \to 0$ (NIS 1.38, close to the upper limit).

6) Estimates on the measured data. The estimates of M2 (Fig. 9) follow the circle smoothly. Their mean radius is 10.21 ($\hat{r} = 10.24$), and their RMS radial deviation from the identified circle is 0.60, against 0.83 for the measurements

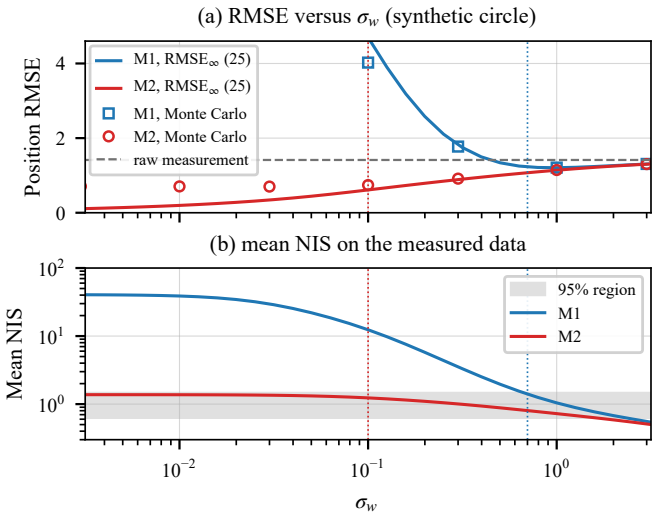


Fig. 12. Part 3: influence of σ_w. (a) Asymptotic RMSE (25) (lines) and Monte Carlo RMSE for $k = 5, \ldots, 24$ (1000 runs, markers); (b) mean NIS on the measured data with the 95% acceptance region. Dotted vertical lines: the chosen values.

($k \geq 5$). The first two estimates equal the measurements because of the diffuse prior, so the outlying first fix $(7.12, 0)$, only 6.8 from the identified centre, cannot be corrected.

7) Limitations and extensions. $\hat{\omega}$ and $\hat{c}$ are batch estimates computed from the whole record. For on-line tracking, ω can be added to the state, giving a coordinated-turn model that requires an extended Kalman filter [7], or several models can be combined in an interacting multiple-model filter [5]. Signal-strength errors are often heavy-tailed, which calls for innovation gating or robust filtering. If a delay is acceptable, a Rauch–Tung–Striebel smoother [11] would also improve the first estimates.

VI. CONCLUSIONS

Kalman filters were designed for the three problems of the project, and every simulated result was explained analytically. In Part 1 the filter converges to an exponential smoother with gain 0.0951, which optimally balances lag against noise and reduces the error variance 10.4-fold. In Part 2 it converges to an α–β tracker with $\alpha = 0.36$ and $\beta = 0.08$, estimates the unmeasured velocity with variance 0.0400, and the bias and variance agree with the Riccati solution once the initial-condition term is taken into account. In Part 3 a constant-velocity filter distorts the circular trajectory, as predicted by its frequency response. A system of two decoupled oscillator-model Kalman filters, whose model is identified from the data, contains the dynamics of the motion. It tracks the circle with unit gain and zero phase, passes the innovation consistency test, and reduces the RMSE by 48%. Across all parts, the Riccati equation predicts the performance before any data are processed, and the filter is only as good as its model.

Three practical lessons follow. First, the gains do not depend on the data, so the Riccati equation can be solved off-line; after roughly 10 to 30 samples every filter of this report

TABLE IV

PART 3, BASELINE DESIGN (CONSTANT-VELOCITY MODEL, $\sigma_w = 0.7$): $K_f(k)$, $K(k)$, $P(k|k)$ AND $P(k+1|k)$ (IDENTICAL FOR THE y- AND z-FILTERS).

	$K_f(k)$		$K(k)$		$P(k\|k)$			$P(k+1\|k)$		
k	K_{f1}	K_{f2}	K_1	K_2	P_{11}	P_{12}	P_{22}	P_{11}	P_{12}	P_{22}
0	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000	1.000×10^5	1.000×10^5	1.000×10^5	1.000×10^5
1	1.0000	1.0000	2.0000	1.0000	1.0000	1.0000	2.1225	5.2449	3.3674	2.6125
2	0.8399	0.5392	1.3791	0.5392	0.8399	0.5392	0.7966	2.8375	1.5809	1.2866
3	0.7394	0.4120	1.1514	0.4120	0.7394	0.4120	0.6354	2.3212	1.2923	1.1254
4	0.6989	0.3891	1.0880	0.3891	0.6989	0.3891	0.6225	2.2222	1.2566	1.1125
5	0.6896	0.3900	1.0796	0.3900	0.6896	0.3900	0.6224	2.2146	1.2574	1.1124
6	0.6889	0.3912	1.0801	0.3912	0.6889	0.3912	0.6206	2.2143	1.2567	1.1106
7	0.6889	0.3910	1.0799	0.3910	0.6889	0.3910	0.6192	2.2126	1.2552	1.1092
8	0.6887	0.3907	1.0794	0.3907	0.6887	0.3907	0.6188	2.2114	1.2545	1.1088
9	0.6886	0.3906	1.0792	0.3906	0.6886	0.3906	0.6187	2.2111	1.2544	1.1087
rows $k = 10, \ldots, 24$ are identical to row 9 (to four decimals)										
∞	0.6886	0.3906	1.0792	0.3906	0.6886	0.3906	0.6187	2.2111	1.2544	1.1087

TABLE V

ESTIMATES $(\hat{x}_y(k|k), \hat{x}_z(k|k))$ OF THE FINAL DESIGN (M2).

k	$\hat{x}_y(k\|k)$	$\hat{x}_z(k\|k)$	k	$\hat{x}_y(k\|k)$	$\hat{x}_z(k\|k)$
0	7.1164	0.0000	13	−9.8231	−2.0081
1	9.6022	3.1398	14	−8.6475	−4.6244
2	9.3576	6.3235	15	−6.8202	−7.5519
3	9.2979	9.4001	16	−4.7121	−9.3090
4	7.3385	10.7797	17	−2.0842	−10.0754
5	4.9609	11.1389	18	0.1801	−10.1006
6	1.8411	10.4447	19	2.8746	−9.7119
7	−0.5852	9.9804	20	6.1244	−8.4780
8	−3.8517	8.3576	21	7.7126	−7.2998
9	−5.8819	6.8753	22	9.2292	−4.9901
10	−8.1642	4.8810	23	9.8727	−2.3049
11	−9.5168	2.6675	24	9.8195	0.0354
12	−9.9544	0.2996			

equals its constant-gain steady-state version (an exponential smoother, an α–β tracker or a tuned oscillator filter). Second, the process-noise covariance R_1 is a design parameter rather than a physical constant: it trades lag against noise and must cover the dynamics that the model omits. Third, the innovation sequence checks a design without any ground truth; here it rejected the inadequate model before any Monte Carlo study was run.

APPENDIX A: ADDITIONAL TABLES FOR PART 3

Table IV gives the gains and covariances of the baseline M1, and Table V the estimates of the final design.

APPENDIX B: SOFTWARE

The MATLAB code is on GitHub (link below the title). It needs no toolbox; its main script run_all.m produces all graphs, statistics and tables of this report. It consists of the following files:

- part1.m–part3.m: Graphs 1–9, Figure 5.1, tables;
- kf.m, kf_gains.m: filter (3)–(8), also with fixed gains;
- kf_steady.m, kf_freq.m: Riccati solution, (10) and $\|H\|_2^2$;
- cv_model.m, osc_model.m, identify_circle.m: (17), (24), Section V-A.

The figures and numbers of this report were produced by an equivalent Python implementation with fixed random seeds. Part 3 contains no random numbers, so the MATLAB results coincide with Tables II, IV and V to all printed digits (checked with GNU Octave 9.4). Parts 1 and 2 use a different random-number generator, so their statistics differ from Table I only within the run-to-run spread. run_all also saves the figures and the Part 3 tables and ends with a self-check against this report. The code follows Algorithm 1, which starts from $\hat{x}(0|-1)$ and $P(0|-1)$.

Algorithm 1. Kalman filter, eqs. (3)–(8)

for $k = 0, 1, \ldots, N$ **do**
  $S(k) \leftarrow CP(k|k-1)C^{\mathsf{T}} + R_2$
  $K_f(k) \leftarrow P(k|k-1)C^{\mathsf{T}}S(k)^{-1}, \;\; K(k) \leftarrow AK_f(k)$
  $e(k) \leftarrow y(k) - C\hat{x}(k|k-1)$
  $\hat{x}(k|k) \leftarrow \hat{x}(k|k-1) + K_f(k)\, e(k)$
  $\hat{x}(k+1|k) \leftarrow A\hat{x}(k|k-1) + Bu(k) + K(k)\, e(k)$
  $P(k|k) \leftarrow P(k|k-1) - K_f(k)\, CP(k|k-1)$
  $P(k+1|k) \leftarrow AP(k|k-1)A^{\mathsf{T}} - K(k)S(k)K^{\mathsf{T}}(k) + R_1$
end for

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